



When hearing the bark helps to identify the dog: Semantically-congruent sounds modulate the identification of masked pictures

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ABSTRACT

We report a series of experiments designed to assess the effect of audiovisual semantic congruency on the identification of visually-presented pictures. Participants made unspeeded identification responses concerning a series of briefly-presented, and then rapidly-masked, pictures. A naturalistic sound was sometimes presented together with the picture at a stimulus onset asynchrony (SOA) that varied between 0 and 533 ms (auditory lagging). The sound could be semantically congruent, semantically incongruent, or else neutral (white noise) with respect to the target picture. The results showed that when the onset of the picture and sound occurred simultaneously, a semantically-congruent sound improved, whereas a semantically-incongruent sound impaired, participants' picture identification performance, as compared to performance in the white-noise control condition. A significant facilitatory effect was also observed at SOAs of around 300 ms, whereas no such semantic congruency effects were observed at the longest interval (533 ms). These results therefore suggest that the neural representations associated with visual and auditory stimuli can interact in a shared semantic system. Furthermore, this crossmodal semantic interaction is not constrained by the need for the strict temporal coincidence of the constituent auditory and visual stimuli. We therefore suggest that audiovisual semantic interactions likely occur in a short-term buffer which rapidly accesses, and temporarily retains, the semantic representations of multisensory stimuli in order to form a coherent multisensory object representation. These results are explained in terms of Potter's (1993) notion of conceptual short-term memory.

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1. Introduction

In everyday life, the majority of perceptual events provide information to multiple sensory modalities simultaneously (e.g., see the chapters in Calvert, Spence, & Stein, 2004). For example, when watching a ball bounce, we normally also hear the synchronized sound of the ball hitting the ground. In such cases, coherent spatial, temporal, and semantic information is conveyed to both the visual and

auditory modalities. The spatiotemporal constraints on crossmodal interactions and multisensory integration have been explored extensively by researchers in recent years (see Spence, 2007, for a review). By contrast, the constraints of semantic congruency on multisensory object identification have, to date, received far less attention. The goal of the present study was therefore to investigate whether the identification of a visually-presented picture can be influenced by the presentation of a semantically-congruent stimulus in another sensory modality (i.e., in this case, audition).

Researchers have been studying the processing of visually-presented pictures and how it can be influenced by the presentation of relevant visual information (such as the vi-

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sual background) for nearly 40 years (e.g., Biederman, 1972; Biederman, Mezzanotte, & Rabinowitz, 1982; Boyce & Pollastek, 1992; Boyce, Pollastek, & Rayner, 1989; Davenport & Potter, 2004; Glaser & Glaser, 1989; Henderson & Hollingworth, 1999; Hollingworth & Henderson, 1998; Intraub, 1981; Intraub, 1984; Palmer, 1975; Potter, 1975, 1976). For example, researchers have shown that the theme of a picture can be detected easily in an on-line detection task while being difficult to recall in a subsequently-presented recognition task (Intraub, 1981; Potter, 1975, 1976). In addition, picture processing can be influenced by the semantic congruency of any supplementary visual information that may happen to be presented at the same time. For example, it has been shown that it is easier to detect or report a visual target when it is presented against a semantically congruent background than when it is presented against a semantically incongruent background instead (Biederman et al., 1982; Boyce et al., 1989; Davenport & Potter, 2004; Palmer, 1975; though see Hollingworth & Henderson, 1998). Similarly, when a word is superimposed on top of a picture, the identification of that picture can either be facilitated by the presentation of a semantically congruent word or else inhibited by the presentation of a semantically incongruent word (Glaser & Dunganhoff, 1984). Furthermore, the identification or recognition of a picture can either be improved, when the subsequently-presented item (either a picture or a word) is semantically congruent with it (Glaser & Glaser, 1989; O'Connor & Potter, 2002), or else impaired when the subsequent item happens to be semantically incongruent (Glaser & Glaser, 1989; Intraub, 1984; Loftus & Ginn, 1984). Potter (1976) labeled the latter phenomenon *conceptual masking*.

These results have three important implications for our understanding of the processes underlying the identification of visually-presented pictures. First, the semantic information associated with a picture can be accessed within 100 ms of stimulus onset, maintained for a further 300 ms, and then either consolidated in memory or else rapidly forgotten (Intraub, 1999; Potter, 1975). Second, irrespective of the nature of the visual stimuli (e.g., picture or word), a shared abstract representation of the semantic information may be accessed if the stimuli are associated with a common concept (Potter & Faulconer, 1975). Therefore, the interactions that are seen at the level of semantic congruency between picture and word presumably rely on a common semantic system (see Glaser & Glaser, 1989). Third, the effects of supplementary visual information on participants' picture identification performance are not constrained by the need for strict temporal coincidence; instead, a subsequently-presented stimulus (e.g., lagging by as much as 350 ms in Loftus & Ginn's, 1984, study) can also influence performance. Potter (1993, 1999) proposed a module, called *conceptual short-term memory* (CSTM) to account for the semantic processing of visual stimuli. Potter thought of CSTM as consisting of a short-term buffer that serves as the interface between perception and memory. Once a visual stimulus has been identified, its meaning is accessed rapidly and maintained temporarily in CSTM. Furthermore, the maintained semantic representations are structured into a coherent context in order to

be consolidated in memory; otherwise, unstructured (usually semantically incongruent) information would be rapidly forgotten (Potter, 1999).

In the present study, we investigated semantic processing for the case of multisensory interactions between visual and auditory stimuli. We hypothesized that a common semantic system can be accessed by a given picture and a semantically-congruent stimulus presented in another sensory modality, such as audition (see Barsalou, 1999; Versace, Labeye, Badard, & Rose, 2009, on this issue). If this hypothesis is correct, the semantic congruency of an auditory stimulus may be expected to influence the identification of a visually-presented picture. Furthermore, we suggest that audiovisual semantic interactions should still occur even when the presentation of the sound lags within a certain interval (around 300 ms) after the picture.

1.1. Previous research on audiovisual semantic congruency effects

Jackson (1953) was perhaps the first researcher to demonstrate the effect of semantic congruency on human information processing in a multisensory setting. He showed that the *spatial ventriloquism effect* (whereby the perceived source of the sound is pulled toward the location of a visual stimulus) occurred over larger spatial disparities for realistic pairs of stimuli (e.g., the whistling sound normally associated with a steam kettle and the sight of a steaming kettle) than for artificial pairings of stimuli (e.g., the sound of a bell and the illumination of a light). That is, it would appear that people can tolerate a larger spatial (or for that matter temporal) separation between congruent (as compared to incongruent or unrelated) pairs of audiovisual stimuli and still bind them. However, this explicit measure of audiovisual integration (namely, pointing to the apparent source of a sound) may unfortunately have resulted from *response bias or the particular decision strategies adopted by participants* (see Bertelson & Aschersleben, 1998; Choe, Welch, Gilford, & Juola, 1975; Vatakis & Spence, 2007). This is probably the reason why, in subsequent studies where the possible influence of such biases on performance have been ruled out, similar effects of audiovisual semantic congruency have not always been observed (e.g., Vatakis, Ghazanfar, & Spence, 2008; Vatakis & Spence, 2008; though see Parise & Spence, 2009; Vatakis & Spence, 2007).

In addition, Koppen, Alsius, and Spence (2008) have recently shown that audiovisual semantic congruency does not modulate the Colavita visual dominance effect (Colavita, 1974). In this paradigm, participants respond to visual and auditory targets accurately when they were presented alone; however, participants sometimes failed to respond to the auditory targets when they were presented together with visual targets (see also Hecht & Reiner, 2009; Spence, Parise, & Chen, submitted for publication). Across three experiments, Koppen et al. demonstrated that the semantic congruency between visual and auditory stimuli (congruent versus incongruent, such as the picture of dog accompanied by the sound of a dog "woofing" or a cat "meowing", respectively) had no effect on the magnitude of the Colavita visual dominance effect. That is, congruent

sounds were detected no more accurately than incongruent sounds on those trials in which they were accompanied by a visual target.

On the other hand, there is some evidence from other experimental paradigms to show that the semantic congruency of visual and auditory stimuli can influence the latency of a participant's behavioral responses. For example, target detection latencies tend to be shorter when a visual target is accompanied by a semantically-congruent sound than when it is accompanied by a semantically-incongruent sound instead (Laurienti, Kraft, Maldjian, Burdette, & Wallace, 2004; Molholm, Ritter, Javitt, & Foxe, 2004; Suied, Bonneel, & Viaud-Delmon, 2009). Performance in visual search tasks accompanied by a sound that happens to be semantically congruent with the visual target has also been shown to be facilitated, as compared to when search displays are accompanied by a target-incongruent sound instead (see Iordanescu, Guzman-Martinez, Grabowecky, & Suzuki, 2008).

It should be noted here that those recent studies that have demonstrated an audiovisual semantic congruency effect have always done so in terms of the speed of participants' behavioral responses. As a consequence, it is unclear whether the audiovisual semantic congruency effect reported in these previous studies should be attributed primarily to a perceptual or to a decisional level effect. In the studies of Laurienti et al. (2004), Molholm et al. (2004), and Suied et al. (2009), the congruent visual and auditory stimuli corresponded to the same response key. It is therefore possible that, when information was provided both visually and auditorily (i.e., in the bimodal condition), the participant's response might have been triggered by the first stimulus that they happened to have perceived (i.e., a statistical facilitation of responses might have been observed, see Raab, 1962). All three of these studies ruled out statistical facilitation as the sole account of their data by comparing the cumulative distribution function of response times (RTs) for the bimodal condition to the predictions of the race model (see Miller, 1982). A further concern though is whether the facilitatory effect reported in the bimodal condition in these studies was truly due to semantic congruency, or alternatively, due to the visual and auditory stimuli both being linked with the same response (i.e., response redundancy, see Experiment 3 in Miller, 1982, on this point).

Suied et al. (2009, in the Appendix of their paper), for example, tried to disentangle semantic congruency from response redundancy by mapping the semantically-incongruent stimuli (e.g., the picture of a phone and the "ribbit" sound made by a frog) to the same response key. However, using this manipulation, the presentation of the telephone ringing sound did not speed-up the participants' responses to the picture of the telephone (i.e., no semantic congruency effect was observed). Evidence supporting the view that response redundancy is critical to the shortened RTs that have been observed in speeded tasks comes from the work of Sinnett, Soto-Faraco, and Spence (2008). In their study, even though the visual and auditory targets were always semantically incongruent (e.g., a picture of "traffic light" paired with the sound of a cat "meowing"), when they were mapped to the same response key, RTs

in the bimodal condition were nevertheless still significantly faster than the predictions of the race model. That is, given that the visual and auditory inputs were processed independently, they were still able to sum together to reach the criterion for initiating a response (this result is in line with the *independent coactivation model*, Miller, 1991). Therefore, one needs to exercise caution before explaining the RT data in speeded tasks in terms of semantic congruency effects since the results are easily confounded with the response redundancy (i.e., a decision-level processing) effect (Santee & Egeth, 1982; Watt, 1991).

Another problem for the above-mentioned studies is that the visual target that participants had to detect, or search for, was clearly defined beforehand (Iordanescu et al., 2008; Koppen et al., 2008; Laurienti et al., 2004; Molholm et al., 2004; Suied et al., 2009). What this means in practice is that the participants in these studies may have been able to accomplish their task simply by recognizing the salient features of the target picture (e.g., four legs in the case of the picture of dog and then inferring the rest). Besides, as the target had been well-defined in advance, a clear goal-directed control mechanism may have been active, which might have guided participants' attention to select the contingent stimuli (e.g., Folk, Remington, & Johnston, 1992). That is, the presentation of the semantically-congruent sound may have been selected for further processing, whereas the semantically-incongruent sound was ignored. The prediction here is that the facilitation elicited by the presentation of a semantically-congruent sound can be observed, whereas the interference elicited by a semantically-incongruent sound should be reduced or even eliminated, consistent with the results of both Iordanescu et al.'s (2008) and Molholm et al.'s (2004) studies.

In the present study, a picture identification task was used to test for the existence of an audiovisual semantic congruency effect. We consider that this task is less sensitive to the influence of response-level processing and goal-directed control than the speeded tasks used in previous research. In each trial, a picture from a corpus of at least 36 pictures was presented briefly and then immediately masked, and the participants had to try and identify the visual target. The accuracy of participants' unspeeded identification responses was recorded as the dependent variable. One of the advantages of this task is that the measurement of the accuracy of participants' performance for briefly-presented targets provides a more sensitive measure of perceptual processing than does the measurement of RTs for targets presented for a longer time (e.g., Norman & Bobrow, 1975; Prinzmetal, McCool, & Park, 2005; Santee & Egeth, 1982). This is because any guessing bias can be estimated and separated from participants' performance, and hence the actual sensitivity of participants' responding can be revealed (see Green & Swets, 1966). Another advantage of using a target identification task is that participants do not know the identity of the upcoming target on each and every trial. The increased uncertainty regarding the identity of the target should therefore help to reduce any attentional selection resulting from goal-directed control that may have been taking place in previous studies (e.g., Yeh & Liao, 2008).

The primary goal of the present study was therefore to provide convergent evidence regarding the influence of audiovisual semantic congruency on the identification of masked visual targets, and further to examine whether the source of any congruency effect was due to the facilitation of performance in the congruent condition, the impairment of performance in the incongruent condition, or to both effects operating simultaneously. Furthermore, in previous studies of audiovisual semantic congruency, the visual and auditory stimuli have always been presented simultaneously (Iordanescu et al., 2008; Koppen et al., 2008; Laurienti et al., 2004; Molholm et al., 2004; Suied et al., 2009). Given that the semantic congruency effect that has been reported within the visual modality can be induced by the presentation of both simultaneous and asynchronous stimuli, it seemed worthwhile to investigate whether the same was true in a crossmodal (i.e., audiovisual) setting as well. The stimulus onset asynchrony (SOA) between the picture and the sound was therefore manipulated; however, because we were primarily interested in the processing of a picture after it had been presented, rather than the semantic priming of a picture by sound (e.g., Schneider, Engel, & Debener, 2008), the auditory stimulus was only presented simultaneously or else delayed with respect to the onset of the picture. In Experiments 1 and 2, the onset of the sound occurred simultaneously with that of the picture. In Experiments 3 and 4, the onset of the sound was delayed by 307 and 533 ms with respect to the picture. In Experiment 5, the SOAs (0, 333, 533 ms) were randomized on a trial-by-trial basis and tested within the same group of participants. Based on the results of semantic congruency studies in visual perception, we predicted that any audiovisual semantic congruency effect should be observed at SOAs shorter than around 300 ms, but not at SOAs longer than 500 ms.

2. Experiment 1

Experiment 1 was designed to investigate whether the semantic congruency (congruent versus incongruent) of a sound would influence the ability of participants to identify briefly-presented line-drawing pictures. We predicted that the presentation of a semantically-congruent sound would facilitate, whereas the presentation of a semantically-incongruent sound would impair, participants' performance relative to a control condition in which the picture was accompanied by a burst of white noise.

2.1. Methods

2.1.1. Participants

Seventeen volunteers (six males, mean age 26 years) took part in this study in exchange for course credit or a five pound (UK Sterling) gift voucher. All of the participants had normal or corrected-to-normal vision and hearing by self-report, and were naïve as to the purpose of the study.

2.1.2. Apparatus and stimuli

The stimuli were presented on a 15-in. CRT monitor (75 Hz refresh rate) controlled by a personal computer. The participants sat at a viewing distance of 58 cm from the monitor in a dimly-lit experimental chamber. Each trial consisted of the presentation of a fixation stimulus (a $0.69^\circ \times 0.69^\circ$ cross), a target picture, and a pattern mask (see Fig. 1A). All of the stimuli were presented in black in the center of a white background. Forty-two outline-drawing figures taken from the Snodgrass and Vanderwart (1980) database were used as the visual target stimuli. The mask consisted of random curves and encompassed an area of $5.32^\circ \times 5.32^\circ$, sufficient to cover the target pictures completely.

The congruent auditory stimuli were chosen to match the pictures: e.g., the sound of a dog's barking was matched to the picture of a dog; the sound of a creaking door was paired with the picture of a door (see Appendix 1). All of the sound files were downloaded from <http://www.findsounds.com> (on 08/08/2008). The sounds were edited to start from the beginning of the sound file and to last for 350 ms with equalized peak amplitudes. The sounds were presented over closed-ear headphones and ranged in loudness from 66 to 76 dB SPL. The white noise in the control condition was presented at 72 dB SPL.

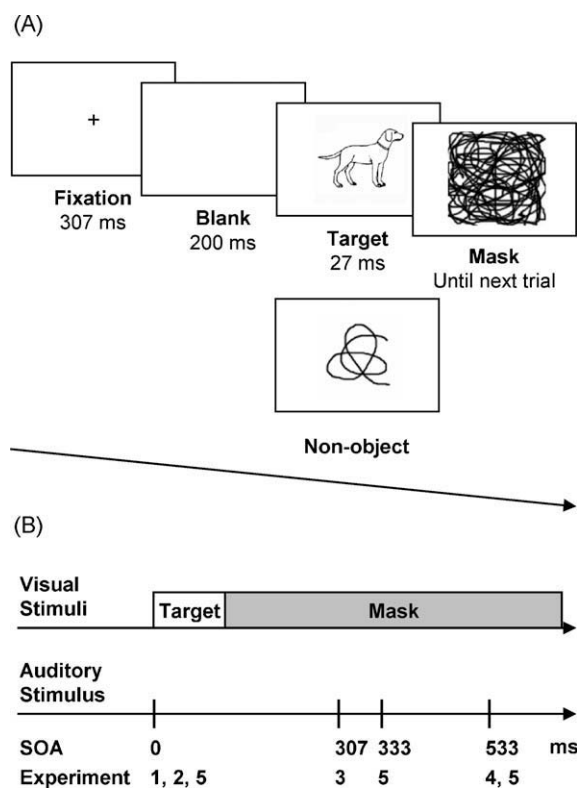


Fig. 1. (A) The sequence of four frames presented in each trial: a fixation stimulus, a blank, a target picture (e.g., dog), and a pattern mask. An example of one of the non-object patterns used in Experiments 2–5 is also shown. (B) A schematic representation of the manipulation of SOAs between the visual target picture and the task-irrelevant accessory sound across the five experiments reported in this study.

2.1.3. Design

On each trial, a picture was accompanied by a sound that was semantically congruent, semantically incongruent, or else neutral (i.e., white noise) with respect to the visual target. Semantic congruency was manipulated as a within-participants factor. In the congruent condition, the sound matched the picture. In the incongruent condition, for fear of any unwanted semantically-related priming effects occurring, the paired sound and picture belonged to different semantic categories. In other words, a picture of an animal was always accompanied by the sound of a tool, or instrument, and so on. In the noise condition, the picture was accompanied by a 350 ms burst of white noise. The onsets of the picture and sound were synchronized.

Three blocks of experimental trials were presented. All 42 pictures were presented once in each block of trials. A third of the pictures were assigned to each of the three sound conditions. The sound condition factor was manipulated according to a Latin square design across the three blocks of trials. The 42 pictures were presented in a completely randomized order within each block of trials, and the order in which the three blocks were presented was counterbalanced across participants.

2.1.4. Procedure

Prior to the start of the experiment, two preview sessions including all of the pictures and sounds were presented. In the visual session, all of the pictures were shown together with their names. Each picture-name pair was presented for 1012 ms. The next picture-name pair was then presented after a further 506 ms blank period. Similarly, the auditory session involved the presentation of all of the sounds and their matching names. Each sound was presented for 350 ms while its name was presented for 1012 ms. The next sound-name pair was then presented after a further 506 ms blank interval. The order of presentation of each picture-name pair or sound-name pair was completely randomized within each session across participants.

In the main experimental session, the participants initiated a trial by pressing the Enter key on the keyboard placed in front of them. The fixation cross was presented for 307 ms, followed by a blank screen for 200 ms. Next, the target picture and the sound were presented together. The picture was presented for 27 ms (two frames at 75 Hz refresh rate) and masked by the pattern mask immediately afterward (Fig. 1B). The mask was used to control the presentation duration of the visual target and the difficulty of the visual task. The participants were instructed to enter the name of the picture on the keyboard. They were also informed that they should respond to the picture rather than the sound, and to guess if they were uncertain.

A practice session consisting of five trials (including all three sound conditions) was conducted prior to the main experiment. The stimuli in the practice session were not used in the main experiment. The participants had to repeat the practice session if they failed to report the name of the picture correctly in at least two of these five trials. The adoption of this criterion resulted in approximately 40% of the participants repeating the practice session.

However, all of the participants achieved this criterion the second time round.

2.2. Results

When participants entered the name of the picture or a word with a similar meaning (e.g., “chicken” for the picture of a rooster) their response was counted as correct. The criterion was held constant across all five experiments reported in this study. The data from one participant (female) was excluded because her overall adjusted accuracy (see below) fell below 10% correct, leaving the data from 16 participants for further analysis.

If participants accidentally entered the name associated with the sound in the congruent condition (i.e., without having seen the visual target), the accuracy of their visual object identification performance would have been artificially overestimated. By contrast, such “auditory” responses would not have increased the accuracy of their responding in either the incongruent or noise conditions. In order to correct for any such guessing bias, the frequency with which participants entered the name of the sound in the incongruent condition was calculated and served as the false alarm (FA) rate for the congruent condition. The accuracy data from the congruent condition and the FA¹ rate were entered into the formula reported by Intraub (1984):

$$(\text{Accuracy} - \text{FA}) / (1 - \text{FA})$$

The adjusted accuracies of participants' visual target identification performance in each condition (see Table 1) were subjected to arc-sine transformation using the equation $X' = 2 * \sin^{-1} \sqrt{X}$ (Kirk, 1995) in order to linearize the percentage data². The transformed data were then submitted to a one-way analysis of variance (ANOVA) with the factor of sound condition. The analysis revealed a significant main effect ($F(2, 30) = 33.44$, $MSE = 0.013$, $p < .0001$, $\eta^2 = 0.69$). A post-hoc Fisher-Hayter test (Kirk, 1995) showed that the adjusted accuracy of participants' performance was significantly higher in the congruent condition than in either the incongruent or noise conditions (both $ps < .01$, $q(30, 2).01 = 3.89$, $MSE = 0.013$). Furthermore, participants' responses were also significantly more accurate in the noise condition than in the incongruent condition ($p < .05$, $q(30, 2).05 = 2.89$). See Fig. 2 for the magnitude of the semantic congruency effects compared to the noise control condition reported in Experiment 1.

¹ We hypothesized that in the picture identification task, responding correctly to the picture should depend on the activation of the appropriate stimulus representation above a certain threshold level, while the criterion set for responding to a target should not change during the course of the experiment. Hence, the response to the identity of the sound should reflect pure guessing rather than misperceiving the picture. Therefore, in line with the high threshold model, we calculated sensitivity according to the equation $(\text{Accuracy} - \text{FA}) / (1 - \text{FA})$ (Green & Swets, 1966; Macmillan & Creelman, 2005). In the congruent condition, accuracy scores should be adjusted by an estimated FA rate; in the incongruent and noise conditions, the FA rate was considered as zero.

² The statistical analysis of the raw data in all five experiments reveal exactly same pattern of results.

Table 1

Mean accuracy (%) and standard error (in parentheses) of participants' responses in Experiments 1–4.

Experiment	Sound condition					Incongruent	Noise	No sound
	Congruent							
	% Correct	FA1	Adjusted 1	FA2	Adjusted 2			
1	70.7 (3.4)	10.7 (2.7)	65.8 (4.3)	–	–	50.9 (5.7)	55.8 (4.8)	–
2	69.7 (4.3)	2.8 (1.2)	69.7 (4.7)	7.0 (2.0)	68.0 (5.0)	61.9 (5.5)	–	63.8 (5.1)
3	76.8 (4.4)	0.9 (0.3)	76.5 (4.6)	5.1 (1.8)	75.9 (4.4)	71.0 (5.1)	72.9 (4.8)	–
4	76.6 (2.7)	0.4 (0.2)	76.5 (2.8)	2.4 (0.9)	75.8 (3.0)	73.2 (2.8)	74.3 (3.4)	–

FA1: The false alarm (FA) rate was estimated from the frequency with which the participants entered the name of the object indicated by the sound as the visual target in the incongruent condition; Adjusted 1: The accuracy of congruent condition was adjusted by FA1; FA2: FA rate was estimated from the frequency with which the participants entered the name of the object indicated by the sound as the visual target in the non-object condition; Adjusted 2: The accuracy of congruent condition was adjusted by FA2.

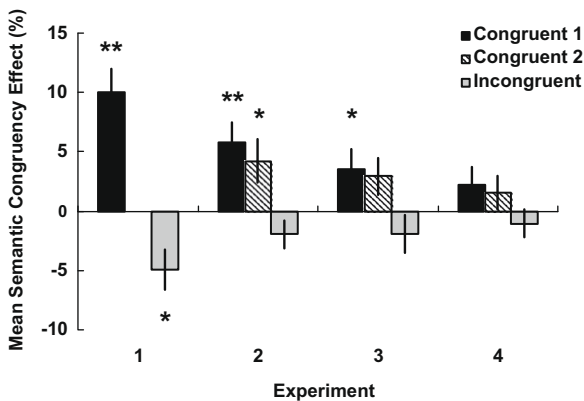


Fig. 2. Graph highlighting the semantic congruency effects (and ± 1 standard error of means) reported in Experiments 1–4. The magnitude of the effect was calculated by subtracting the accuracy of participants' performance in the semantically congruent condition (or semantically incongruent condition) from that in the noise condition (Experiments 1, 3, and 4) or no-sound condition (Experiment 2). Congruent 1 and 2 refers to the accuracy adjusted from FA1 and FA2, respectively (see Table 1). Positive values indicate the improvement in the semantically congruent condition, whereas negative values indicate the impairment in the semantically incongruent condition relative to the noise (or no-sound) condition. * denotes a significant difference ($p < .05$; ** $p < .01$).

2.3. Discussion

The results of Experiment 1 highlighted the existence of an audiovisual semantic congruency effect: That is, the participants were better able to identify the masked pictures when a semantically-congruent sound was presented at the same time than when a semantically-incongruent sound was presented instead. This result is consistent with the results of other studies in which the presentation of a semantically-congruent sound has been shown to facilitate the ability of participants to rapidly detect, or search for, a given visual target (Iordanescu et al., 2008; Laurienti et al., 2004; Molholm et al., 2004; Suied et al., 2009). Taken together, it would therefore appear that audiovisual semantic congruency effects can affect both the speed and accuracy of participants' behavioral performance.

The results of Experiment 1 also showed that the presentation of a semantically-congruent sound facilitated, whereas the presentation of a semantically-incongruent sound impaired, participants' picture identification perfor-

mance when compared to performance in a control condition involving the presentation of white noise. Similar results (in terms of the RT data) were reported by both Laurienti et al. (2004)³ and Suied et al. (2009), while only a facilitatory effect in the semantically congruent condition was observed in the studies of Iordanescu et al. (2008) and Molholm et al. (2004). This inconsistency in terms of whether or not impaired performance is observed in the semantically-incongruent condition may be attributable to the specific control conditions used in the various studies; in particular, the control conditions were unimodal (i.e., respond to either a visual or auditory target which is presented alone).

In our next experiment, we therefore decided to investigate whether both facilitatory and inhibitory effects of audiovisual semantic congruency would still be observed when our participants' performance in the congruent and incongruent trials was compared to that in just such a unimodal control condition, here involving picture identification in the absence of any sound. The prediction regarding participants' performance in this condition is not entirely obvious: On the one hand, the no-sound condition might be expected to give rise to performance that is no different from that seen in the incongruent conditions, as in the studies of Iordanescu et al. (2008) and Molholm et al. (2004). Alternatively, however, the no-sound condition might also be expected to give rise to result lying in-between that seen in the incongruent and congruent conditions, just as reported in the noise condition of Experiment 1, given that in neither case is any useful semantic information provided via the auditory modality. A final possibility is that when the visual picture is presented in isolation, performance is even better than in the congruent condition, because participants can fully focus their attentional resources on the processing of the target picture. Previous research has documented the

³ In Laurienti et al.'s (2004) study, RTs in the incongruent condition were slower than those reported in the unimodal (both visual and auditory) conditions. However, the stimuli used in the incongruent condition consisted of the pairing of a target in one modality with a task-irrelevant stimulus in the other modality. A similar manipulation was actually called the *single target with distractor* condition in Miller's (1991) study. The RT for this condition in Miller's study was even slower than that observed in the *incongruent redundant targets* condition in which the stimuli in the pair were both task-relevant but incongruent for the visual and auditory modalities.

deterioration of visual target identification performance seen when additional information is presented within either the same modality (e.g., Davenport & Potter, 2004; Kahneman, Treisman, & Burkell, 1983) or else in a different modality (e.g., Shimojo, Watanabe, & Scheier, 2001; Spence, Ranson, & Driver, 2000). Researchers have attributed this performance decrement to the need for participants to engage their limited attentional resources in order to filter out the irrelevant non-target information (Lavie, 2005), or even to the fact that the non-target information may itself serve to divert attention away from the target (see Dalton & Spence, 2007).

3. Experiment 2

There were two principal goals in Experiment 2: First, to try and replicate the audiovisual semantic congruency effect in the masked picture identification task reported in Experiment 1. The second aim was to compare the semantic congruency effect elicited by the presentation of a sound to a unimodal (i.e., visual only) baseline condition.

3.1. Methods

3.1.1. Participants

Eighteen naïve volunteers (two males, mean age 24 years) from the same pool of volunteers as in Experiment 1 took part in this experiment.

3.1.2. Apparatus, stimuli, design, and procedure

The three levels of the sound condition factor (congruent, incongruent, and no-sound) were manipulated on a within-participants basis. The former two conditions were identical to those presented in Experiment 1. In the no-sound condition, the visual stimulus was presented in the absence of any auditory stimulation. A novel condition, in which a non-object picture was presented, was also included in this experiment. In this non-object condition, 14 non-object visual patterns (see Fig. 1A for an example) were accompanied by naturalistic sounds associated with those pictures used in the no-sound condition. The participants were instructed to respond by pressing the 'Space bar' in the non-object trials. The inclusion of these non-object trials was designed to reduce the likelihood that the participants would respond by trying to guess the name of the sound in those trials in which they were unable to identify the picture. It also provided us with a second estimate of the FA rate, namely, the frequency with which the participants entered the name of the sound when an object picture was not presented. Fifty-six trials were presented in each block of trials, with the same number of trials being presented in the congruent, incongruent, no-sound, and non-object conditions, respectively. All of the other details were the same as in Experiment 1.

3.2. Results

The data from two participants were excluded from the data analysis: They entered the name of the sound in 64% and 45% of trials in the non-object condition, respectively

(this compares to an average of only 7% for the other participants). Given that these participants failed to follow the instructions not to respond to the auditory stimulus, their data were discarded, leaving 16 participants for further analysis (see Table 1 for the results).

The adjusted accuracy data for the three sound conditions was submitted to a one-way ANOVA. This analysis revealed a significant main effect ($F_1(2, 30) = 10.13$, $MSE_1 = 0.012$, $p_1 < .0005$, $\eta^2_1 = 0.40$; $F_2(2, 30) = 5.96$, $MSE_2 = 0.012$, $p_2 < .01$, $\eta^2_2 = 0.28$)⁴. A *post-hoc* Fisher–Hayter test showed that the adjusted accuracy of participants' performance was significantly higher in the congruent condition than in either the incongruent (both $ps < .01$, $q(30, 2).01 = 3.89$, $MSE_1 = 0.012$; $MSE_2 = 0.012$) or no-sound conditions ($p_1 < .01$; $p_2 < .05$; $q(30, 2).05 = 2.89$). The difference in accuracy between the latter two conditions failed to reach statistical significance.

3.3. Discussion

The results of Experiment 2 once again highlighted the fact that it was easier for participants to identify masked line drawings when they were presented together with semantically-congruent sounds than when they were accompanied by semantically-incongruent sounds, thus replicating the audiovisual semantic congruency effect first reported in Experiment 1. Participants' performance in the no-sound condition was not significantly different from that reported in the incongruent condition, which is consistent with the findings of previous research (see Iordanescu et al., 2008; Molholm et al., 2004). A possible explanation for this finding is that in the no-sound condition, the participants were simply less alert than in those trials in which a sound was presented (e.g., Nickerson, 1973; Robertson, Mattingley, Rorden, & Driver, 1998; Spence & Driver, 1997). On the other hand, this result also implies that the presentation of an auditory stimulus did not in-and-of-itself interfere with the participants' processing of the visual target. It has been argued elsewhere that crossmodal interference is typically weaker than intramodal interference, at least at the earliest stages of human information processing (e.g., Alais, Morrone, & Burr, 2006; Laurienti et al., 2004). These results are consistent with previous claims that perceptual/attentional processing resources may be, at least to a certain extent, independent for different sensory modalities (Duncan, Martens, & Ward, 1997; Hancock, Oron-Gilad, & Szalma, 2007; Rees, Frith, & Lavie, 2001; Soto-Faraco & Spence, 2002; Wickens, 1980, 1984; though see Arnell & Larson, 2002; Santangelo, Belardinelli, & Spence, 2007; Spence, Nicholls, & Driver, 2001). By contrast, central processing resources are thought to constitute a relatively unitary pool regardless of the sensory modality of input

⁴ The FA rate was first estimated from the frequency with which the participants entered the name of the sound in the *incongruent* condition (i.e., using the same procedure as in Experiment 1). These statistical values are denoted by F_1 etc. An alternative estimation of the FA rate came from the frequency with which the participants entered the name of the sound in the *non-object* condition. These statistical values are denoted by F_2 etc. Generally speaking, the same pattern of results was obtained when the FA rates were estimated from the incongruent and the non-object conditions. Exceptions are specified in the text.

information (e.g., as when loading working memory or executive control functions; Brand-D'Abrescia & Lavie, 2008; Klemen, Buchel, Buhler, Menz, & Rose, *in press*; Potter, Chun, Banks, & Muckenhaupt, 1998; Spence & Soto-Faraco, *in press*; Treisman & Davies, 1973). Semantic processing would thus appear to represent a case of central processing in which the information from the different senses is pooled and hence can interact crossmodally.

It is important to note that while the pattern mask used in Experiments 1 and 2 should have blocked visual feature processing, it should not have interfered with the conceptual processing of the picture by participants (see Kiefer, 2002; Marcel, 1983a). Therefore, even though the picture was only presented for 27 ms and then masked, and the identification of the sound may well have occurred long after the picture had been masked, the semantic information contained in the auditory and visual stimuli nevertheless still interacted. Given that conceptual processing takes several hundred milliseconds to complete (Potter, 1975), it is presumably possible for the later-identified sound to influence the processing of the picture in a retrospective manner. To further test this hypothesis, the onset time of the sound was delayed with respect to that of the picture in the next experiment.

4. Experiment 3

In this experiment, the onset of the sound occurred 307 ms after that of the picture. The prediction was that the semantically-congruent sound would still likely facilitate, while the semantically-incongruent sound would still inhibit, participants' picture identification performance, as compared to the performance seen in the control noise condition.

4.1. Methods

4.1.1. Participants

Sixteen naïve volunteers (three males, mean age of 18 years) from the same pool of volunteers took part in this experiment.

4.1.2. Stimuli, apparatus, design, and procedure

Three levels of the sound condition factor (congruent, incongruent, and noise) were manipulated on a within-participants basis, just as in Experiment 1. The non-object trials, first introduced in Experiment 2, were included here as well. The sound was presented 280 ms after the onset of the pattern mask (i.e., 307 ms after the onset of the picture, see Fig. 1B). All other details were the same as for Experiment 2.

4.2. Results and discussion

The adjusted accuracy of participants' picture identification performance is shown in Table 1. The main effect of sound condition was again significant ($F_1(2, 30) = 6.10$, $MSE_1 = 0.013$, $p_1 < .01$, $\eta^2_1 = 0.29$; $F_2(2, 30) = 4.30$, $MSE_2 = 0.013$, $p_2 < .05$, $\eta^2_2 = 0.22$). A *post-hoc* Fisher–Hayter test revealed that the adjusted accuracy of participants' performance was also significantly higher in the congruent

condition than in the incongruent condition ($ps < .01$; $q(30, 2).01 = 3.89$, $MSE_1 = 0.013$; $MSE_2 = 0.013$). The adjusted accuracy was significantly higher in the congruent condition than in the noise condition when the FA rate was estimated from the incongruent condition ($p_1 < .05$; $q(30, 2).01 = 2.89$). The difference between the accuracy of participants' performance in the incongruent and noise conditions did not reach statistical significance.

Although the onset of the sound was delayed by 307 ms with respect to that of the picture in this experiment, our results nevertheless still show the facilitatory effect of the semantically-congruent sound on participants' performance. However, the interfering effect of the semantically-incongruent sound was numerically much smaller in this experiment and failed to reach statistical significance. These results therefore imply that the facilitatory and interfering effects elicited by the presentation of semantically congruent versus incongruent sounds likely have somewhat different time-courses.

Contrary to the results of previous unimodal visual studies showing that a subsequently-presented semantically incongruent picture can induce the conceptual masking effect (Intraub, 1984; Loftus & Ginn, 1984), a subsequently-presented auditory stimulus cannot. In the present experiment, the SOA between the picture and sound was 307 ms (note that the SOA between the target and mask pictures was 112 ms in Intraub's, 1984, study and 350 ms in Loftus and Ginn's, 1984, study), which should have been within the time window shown to give rise to conceptual masking. However, it is important to bear in mind that it may take longer to access the meaning of a sound than to access the meaning of a picture, since the meaning of the auditory signal evolves over time, whereas all of the visual features in the picture are presented at once (e.g., Ballas, 1993). The fact that the meaning of the sound and picture may be accessed at different points in time makes it difficult to compare the conceptual masking effects induced by subsequently-presented sounds versus subsequently-presented pictures. Given this concern, we conducted an additional experiment with a shorter SOA between the target picture and the sound (147 ms). However, the results still showed that only the facilitatory effect in the congruent condition was observed, but not the interference effect in the incongruent condition⁵. This result therefore decreases the likelihood that access to the meaning of the sound might simply have occurred outside of the time window in which conceptual masking effects can be demonstrated.

A second possibility here is that the pictures used by Intraub (1984) and Loftus and Ginn (1984) were more

⁵ An additional group of 16 naïve volunteers (three males, mean age of 18 years) took part in this experiment. The only difference from Experiment 3 was that the SOA between the picture and the sound was reduced to 147 ms. The results revealed a significant main effect of sound condition ($F_1(2, 30) = 9.31$, $MSE_1 = 0.022$, $p_1 < .001$, $\eta^2_1 = 0.38$; $F_2(2, 30) = 7.35$, $MSE_2 = 0.021$, $p_2 < .005$, $\eta^2_2 = 0.33$). A *post-hoc* Fisher–Hayter test revealed that the adjusted accuracy of participants' performance was significantly higher in the congruent condition than in the incongruent condition ($ps < .01$; $q(30, 2).01 = 3.89$, $MSE_1 = 0.022$; $MSE_2 = 0.021$) and the noise condition ($p_1 < .01$; $p_2 < .01$, $q(30, 2).01 = 2.89$). The difference between the accuracy of participants' performance in the incongruent and noise conditions did not reach statistical significance.

complicated (e.g., they included more color and detail) than the black-and-white line-drawings used in the present study. If it is true that complex pictures require more time in order to be consolidated, then the conceptual masking effect would be expected to be more pronounced for complex pictures than for simple pictures. A final possibility here is that intramodal competition is not only stronger than crossmodal competition at the early perceptual level of information processing, but also at the later semantic level. It is likely that the incoming information will overwrite any representations that have been presented earlier within the same modality in order to update for the latest changes taking place in the environment, whilst it may only serve to provide supplementary information to the old representations in another modality. However, this account depends *a priori* on the modality source of the sensory input still being represented at the semantic level of information processing (e.g., Barsalou, 1999; McCarthy & Warrington, 1988; Stanfield & Zwaan, 2001; though see Caramazza & Mahon, 2003; Pylyshyn, 1984).

We therefore wondered whether the audiovisual semantic congruency effect reported in the three experiments described so far would disappear if the SOA between picture and sound were to be increased still further. Addressing this question should help us to rule out the possibility that the facilitatory effect simply represents some form of decisional-level facilitation effect. That is, the presentation of the semantically-congruent sound might simply have served to reduce any uncertainty that the participants may have had when trying to identify the picture. For example, if participants only identified the picture as a four-legged animal, say, and then heard a barking dog, they may have entered “dog” as their answer. In order to rule out this uncertainty-reduction hypothesis as a possible explanation for our findings, we prolonged the SOA to 533 ms in the next experiment. According to the uncertainty-reduction hypothesis, semantically-congruent sounds should always facilitate participants' performance providing that they appear prior to a participant's response (i.e., regardless of the SOA). On the other hand, if the sound arrived after the semantic representation of the picture has either been consolidated, or else had decayed (see Potter, 1999), no such facilitation should result from the presentation of a semantically congruent sound.

5. Experiment 4

In this experiment, the onset of the sound was delayed by 533 ms with respect to that of the picture. We predicted that the semantic congruency of the sound would no longer have any influence on participants' picture identification performance.

5.1. Methods

5.1.1. Participants

Sixteen naïve volunteers (three males, mean age of 19 years) from the same pool of volunteers took part in this experiment.

5.1.2. Stimuli, apparatus, design, and procedure

Three levels of the sound condition factor (congruent, incongruent, and noise) were manipulated on a within-participants basis. The sound was presented 506 ms after the onset of the pattern mask (i.e., 533 ms after the onset of the picture, see Fig. 1B). All of the other details were the same as in Experiment 3.

5.2. Results and discussion

The adjusted accuracy of participants' picture identification performance is shown in Table 1. There was no main effect of sound condition ($F_1(2, 30) = 2.36$, $MSE_1 = 0.011$, $p_1 = .11$, $\eta_1^2 = 0.14$; $F_2(2, 30) = 1.67$, $MSE_2 = 0.010$, $p_2 = .22$, $\eta_2^2 = 0.10$). This result is consistent with our prediction that when the onset of a sound is delayed sufficiently (i.e., by 533 ms in this experiment), the semantic congruency of the sound no longer influences participants' picture identification performance.

The null effect of the semantic congruency manipulation in the present experiment did not simply result from participants initiating their manual responses before they had actually perceived the sound, as evidenced by the following observation: Analysis of the RT (the time to enter the first letter of answer) data recorded for four of our participants revealed that the participants started to respond 434 ms (average of the fastest RT) after the presentation of the sound (mean was 1924 ms, standard deviation was 1035 ms). Hence, the sound would already have been presented in its entirety by the time that the participants initiated their response. Given that the data still showed that performance in the congruent condition was slightly better than in the incongruent condition, the semantically-congruent sound may have reduced the uncertainty of participants' responses somewhat. However, it should be noted that this form of uncertainty reduction was clearly not the determining factor modulating the facilitation seen in the congruent condition, since the difference from the incongruent condition failed to reach statistical significance.

Before outlining a possible explanation regarding the mechanisms underlying the audiovisual semantic congruency effect, two further possibilities need to be ruled out. First, it could be argued that participants may have adapted to the constant SOA of 307 ms presented in Experiment 3. That is, it has recently been shown that people can (to a certain extent) recalibrate to asynchronously-presented audiovisual stimuli (i.e., and bind them into a single multisensory event; see Fujisaki, Shimojo, Kashino, & Nishida, 2004; Navarra et al., 2005). However, the 533 ms SOA used in Experiment 4 likely fell well outside of the temporal window for audiovisual recalibration, so the visual and auditory stimuli would not have been bound together (Fujisaki et al., 2004). Second, different participants took part in Experiments 1–4, so we thought it prudent to examine the time course of the audiovisual semantic congruency effect within the same group of participants. Therefore, in our last experiment, the three SOAs (0, 333, and 533 ms) were randomized on a trial-by-trial basis and tested on a within-participants basis in order to replicate the results in Experiments 1, 3, and 4.

6. Experiment 5

In this experiment, we attempted to replicate the results of the previous experiments but now using a within-participants experimental design. The SOAs (0, 333, or 533 ms, see Fig. 1B) were randomized on a trial-by-trial basis. This randomization procedure would have ensured that the participants were unable to predict (and, more importantly, to adapt to) the constantly-varying SOA between the picture and the sound.

6.1. Methods

6.1.1. Participants

Twenty-one naïve volunteers (nine males, mean age of 21 years) from the same pool of volunteers took part in this experiment.

6.1.2. Stimuli, apparatus, design, and procedure

Six of the pictures used in previous experiments were excluded from this, our final, experiment. They were three pictures with the highest accuracy (bottle, car, and flute, with accuracies of 98%, 92%, and 92%, respectively) and the three with the lowest accuracy (bee, fly, and grasshopper, with accuracies of 4%, 15%, and 15%, respectively) in Experiment 1. Hence, a total of 36 pictures were used in this experiment.

Two factors, sound condition (congruent, incongruent, or noise) and SOA (0, 333, or 533 ms) were manipulated. Each sound x SOA condition was tested with each one of the 36 pictures. These trials were divided into three blocks, each having balanced sound conditions and SOAs. In each block, 12 non-object patterns were presented three times accompanied by three different naturalistic sounds on each occasion, while the SOA of each non-object-and-sound pair was counterbalanced across blocks. There were 144 trials in each block and the order of presentation of these trials was randomized within each block. The order of presentation of these three blocks was counterbalanced across participants.

The procedure was modified in order to be sure that the sound had been presented in its entirety before the participants made their response. A visual cue “Answer” was presented on the computer monitor 998 ms after the offset of target picture. The participants were instructed to withhold their response until this cue had been presented. They

were warned that their response would not be recorded if they did not follow this instruction.

6.2. Results

Three participants (two males) were excluded from the data analysis: Two of these participant responded at above 95% correct, indicating that the task may have been too easy for them, while the other participant did not wait until the cue to respond, and so a number of his/her responses were not recorded. Eighteen participants therefore remained for further analysis (see Table 2 for the results). There was a main effect of sound condition ($F_1(2, 34) = 23.90$, $MSE_1 = 0.024$, $p_1 < .0001$, $\eta_1^2 = 0.58$; $F_2(2, 34) = 20.73$, $MSE_2 = 0.020$, $p_2 < .0001$, $\eta_2^2 = 0.55$). A *post-hoc* Fisher–Hayter test showed that the adjusted accuracy of participants’ performance was significantly higher in the congruent condition than in either the incongruent sound or noise conditions (all $ps < .01$; $q(34, 2).01 = 3.86$, $MSE_1 = 0.024$; $MSE_2 = 0.020$). There was also a significant interaction between sound condition and SOA ($F_1(4, 68) = 2.90$, $MSE_1 = 0.023$, $p_1 < .05$, $\eta_1^2 = 0.15$; $F_2(4, 68) = 2.80$, $MSE_2 = 0.024$, $p_2 < .05$, $\eta_2^2 = 0.14$). The simple main effect showed that the sound condition factor had a significant effect on performance at the 0 ms SOA ($F_1(2, 102) = 15.61$, $MSE_1 = 0.023$, $p_1 < .0001$, $\eta_1^2 = 0.19$; $F_2(2, 102) = 12.68$, $MSE_2 = 0.023$, $p_2 < .0001$, $\eta_2^2 = 0.17$), and at the 333 ms SOA ($F_1(2, 102) = 12.15$, $p_1 < .0001$, $\eta_1^2 = 0.15$; $F_2(2, 102) = 10.07$, $p_2 < .0005$, $\eta_2^2 = 0.13$), but crucially not at the 533 ms SOA ($F_1(2, 102) = 2.57$, $p_1 = .08$, $\eta_1^2 = 0.03$; $F_2(2, 102) = 1.57$, $p_2 = .23$, $\eta_2^2 = 0.02$).

At the 0 ms SOA, a *post-hoc* Fisher–Hayter test revealed that the adjusted accuracy of participants’ performance was significantly higher in the congruent condition than in either the incongruent or noise conditions (all $ps < .01$; $q(102, 2).01 = 3.72$, $MSE_1 = 0.023$; $MSE_2 = 0.023$), and the accuracy of participants’ performance in the noise condition was significantly higher than in the incongruent condition (both $ps < .05$; $q(102, 2).05 = 2.81$). At the 333 ms SOA, a *post-hoc* Fisher–Hayter test showed that the adjusted accuracy of participants’ performance was significantly higher in the congruent condition than in either the incongruent or noise conditions (all $ps < .01$), while the difference between accuracy levels reported in the noise and incongruent conditions did not reach statistical significance (see Fig. 3).

Table 2

Mean accuracy (%) and standard error (in parentheses) of participants’ responses in Experiment 5.

SOA (ms)	Sound condition					
	Congruent					Incongruent
	% Correct	FA1	Adjusted 1	FA2	Adjusted 2	
0	79.3 (3.5)	2.3 (0.8)	78.8 (3.5)	6.5 (2.0)	77.7 (3.6)	67.4 (4.3)
333	80.6 (2.7)	1.4 (0.7)	80.3 (2.7)	4.8 (1.4)	79.5 (2.8)	70.5 (3.7)
533	76.7 (3.0)	0.5 (0.3)	76.6 (3.0)	4.2 (1.4)	75.4 (3.2)	73.6 (3.6)

FA1: The false alarm (FA) rate was estimated from the frequency with which the participants entered the name of the object indicated by the sound as the visual target in the incongruent condition; Adjusted 1: The accuracy of congruent condition was adjusted by FA1; FA2: FA rate was estimated from the frequency with which the participants entered the name of the object indicated by the sound as the visual target in the non-object condition; Adjusted 2: The accuracy of congruent condition was adjusted by FA2.

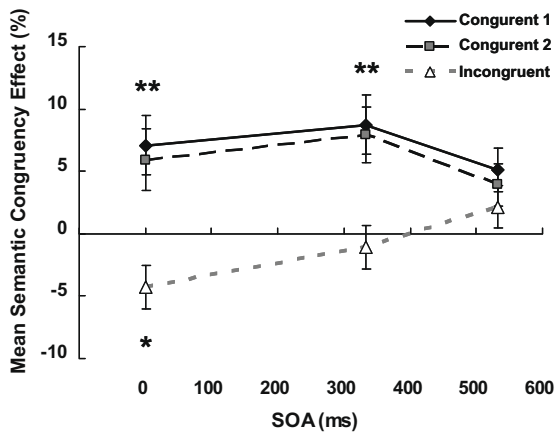


Fig. 3. Graph highlighting the semantic congruency effects (and ± 1 standard error of means) reported in Experiment 5. The magnitude of the effect was calculated by subtracting the accuracy of participants' performance in the semantically congruent condition (or semantically incongruent condition) from that in the noise condition. The congruent 1 and 2 was the accuracy adjusted from FA1 and FA2, respectively (see Table 2). Positive values indicate the improvement in the semantically congruent condition, whereas negative values indicate the impairment in the semantically incongruent condition relative to the noise condition. * denotes a significant difference ($p < .05$; ** $p < .01$).

6.3. Discussion

The results of Experiment 5 therefore replicated those of Experiments 1, 3, and 4. When the onset of the sound coincided with that of the picture, the presentation of a semantically-congruent sound facilitated participants' performance whereas a semantically-incongruent sound impaired performance (as compared to that seen in the noise condition) on the picture identification task. When the onset of the sound was delayed by 333 ms with respect to that of the picture, the facilitatory effect of the congruent sound remained intact. However, when the sound was delayed by 533 ms, the semantic congruency of the sound no longer influenced participants' picture identification performance.

Three main conclusions can be drawn from the results of this final experiment: First, the audiovisual semantic congruency effects observed across the two shortest SOAs (0 and 333 ms), and the null effect of semantic congruency reported at the longest SOA (533 ms) are robust, no matter whether tested on a between- or within-participants basis. Second, since the manipulation of the SOA between picture and sound was randomized on a trial-by-trial basis in this experiment, our participants would have had little opportunity to adapt (and recalibrate) to the changing intervals between the picture and sound. The difference between the results in the 333 and 533 ms SOAs in the present experiment therefore cannot be attributed to any form of audiovisual temporal recalibration. Third, even though the participants in Experiment 5 had to respond about 1000 ms after the offset of the picture, the semantic congruency of the sound nevertheless still influenced their performance at the 0 ms and 333 ms SOAs, but not when the SOA was increased to 533 ms (for a similar result at

the 0 ms SOA condition, see Yuval-Greenberg & Deouell, 2009). This result therefore suggests that the representation of the picture was undergoing some form of dynamic processing across these three intervals, and that the meaning of the sound only influenced the processing of the picture within a certain time window (which was smaller than 533 ms).

7. General discussion

The results of the five experiments reported in the present study demonstrate that the semantic congruency of a sound can influence participants' identification of a masked line-drawing picture. Specifically, when the picture and the sound were presented at the same time, participants' picture identification performance was facilitated by the presentation of the semantically-congruent sound, as compared to when the picture was presented together with a burst of white noise, or in the absence of any sound (see Experiments 1, 2, and 5). On the other hand, performance was impaired when a semantically-incongruent sound was presented at the same time as the picture, as compared to when a burst of white noise was presented instead (Experiments 1 and 5). When the onset of the sound was delayed by a little more than 300 ms with respect to that of the picture, facilitation by the semantically-congruent sound was still observed (see Experiments 3 and 5). However, there was no longer any evidence that audiovisual semantic congruency effects influenced performance once the sound had been delayed by 533 ms (see Experiments 4 and 5).

These results therefore provide the first empirical evidence that audiovisual semantic congruency effects can influence participants' performance on a masked visual picture identification task. Given that the *accuracy* of participants' unspeeded responses was adjusted to take account of their estimated guessing bias, this index should closely approximate the true sensitivity of participants' picture identification abilities. It is easier to account for the congruency effects in terms of a genuine perceptual effect (rather than in terms of a response effect; see Suied et al., 2009) given our use of a sensitivity measure rather than the RT measure used in several previous studies (see Davenport & Potter, 2004; McDonald, Teder-Sälejärvi, & Hillyard, 2000; Prinzmetal et al., 2005; Santee & Egeth, 1982, on this issue). The audiovisual semantic congruency effect observed in the present study therefore suggests that the abstract semantic representations which are associated with the visual and auditory stimuli are processed by the same semantic system. Furthermore, the novel observation reported here was that not only did the semantically-congruent sounds facilitate participants' picture identification performance (when presented simultaneously), but those that had been delayed as much as 333 ms did so as well. On the other hand, the semantically-incongruent sound only impaired participants' performance when its onset coincided with that of the picture.

The observation of a crossmodal semantic congruency effect in the present study implies that the semantic systems for different senses should not be considered as being

completely independent (see McCarthy & Warrington, 1988). On the other hand, since we also suggested that the modality source of sensory input is still represented at the semantic level (see the Discussion of Experiment 3), the semantic system cannot be considered as being completely amodal either (e.g., Pylyshyn, 1984). Some form of intermediate representation between these two extreme hypotheses may therefore be more plausible in terms of accounting for the semantic system in humans, such as a analogue system that retains access to modality information (Barsalou, 1999), a interconnected multimodal network (Shallice, 1988), or else a hierarchical modal-to-amodal system (Plaut, 2002).

It could be argued that our participants may have decided strategically to attend to the identity of the sound, since the auditory stimulus provided potentially useful information with regard to the identity of the target. Note that the auditory and visual stimuli were matched on one-third (Experiment 1) or one-fourth (Experiments 2–5) of all trials. Thus, the sound correctly indicated the identity of the target picture at a level that is far higher than chance. Consequently, it is hard to assert unequivocally that the meaning of the sound was processed “automatically” by participants in the present study (cf. Schneider, Dumais, & Shiffrin, 1984). On the other hand, given that the information from different sensory modalities often helps to identify an object, as in the ball bouncing example discussed in the Introduction, the strategy of attending to multisensory inputs is certainly appropriate in daily life (and not just in an experimental setting). It would therefore seem that, even though the auditory input may first undergo some form of gating by selective attention at an early stage of information processing (e.g., Treisman & Riley, 1969), once it has been selected and processed, its meaning may then unavoidably interact with any relevant visual information (see Lavie, 2005; Lavie, Hirst, de Fockert, & Viding, 2004; Treisman & Davies, 1973; Yuval-Greenberg & Deouell, 2009).

We believe that Potter's idea regarding CSTM (see Potter, 1993, 1999) may provide a useful theoretical framework with which to explain the audiovisual interactions in human information processing reported here. Potter suggested that the meaning of a picture is accessed rapidly (within the first 100 ms or so), and the activated semantic representation is then retained in CSTM for a further 300 ms. At the retention stage, the semantic representation of the picture, as well as other relevant semantic information, are then consolidated in order to be encoded into memory (for a review, see Intraub, 1999). The results of the present study therefore suggest that the processing of a visually-presented picture prior to, and in, CSTM can also be influenced by the meaning of any sound that happens to be presented at around the same time. While a picture is being processed and a sound perceived, access to the meaning of that picture can either be enhanced or interfered with as a function of the meaning of the sound being either congruent or incongruent with it (i.e., at the SOA of 0 ms). When the representation of a picture passes on to the retention stage (i.e., once its meaning has been accessed), the congruent sound can only help with the consolidation of the

picture in CSTM (see also Murray et al., 2004). It is important to note that, at the retention stage, those activated semantic representations referring to the input information are dynamically structured in order to achieve the participants' current task goal (Potter, 1999; Potter, Stiefbold, & Moryadas, 1998). When this goal happens to involve identifying a visually-presented picture, as in the present study, the presentation of a semantically-congruent sound would therefore provide coherent information, and so enhance the consolidation of the picture representation. On the other hand, the presentation of the semantically-incongruent sound was certainly not consistent with the representation of the target picture (e.g., the incongruent sound always belonged to another category in the present study), and hence, being useless to the task at hand, the sound should rapidly have been forgotten (see Potter, 1999). We suggest that the results of the present study therefore not only provide evidence concerning audiovisual interactions taking place at the semantic level but, in turn, suggest that CSTM is *not* a unimodal visual structure.

Potter's (1993, 1999) CSTM construct therefore provides an explanation for why the representation of a picture would be influenced by a simultaneously- or subsequently-presented (by up to around 300 ms) sound in the semantic domain. That is, the mechanism of CSTM is not constrained by the need for strict temporal coincidence between the interacting stimuli. Indeed, this interval is more than sufficient for participants to correctly judge the order of (or perceive as asynchronous) visual and auditory stimuli (e.g., Stone et al., 2001; Vatakis & Spence, 2008; Zampini, Guest, Shore, & Spence, 2005). Soto-Faraco and Alsius (2009) recently demonstrated that the McGurk effect (McGurk & MacDonald, 1976) occurs over a range of asynchronies (from sound leading by 230 ms to sound lagging by 480 ms) that is much wider than the temporal window in which the stimuli are perceived as being simultaneous (from sound leading by 70 ms to sound lagging by 234 ms). That is, the McGurk effect also occurs at SOAs at which participants can accurately judge that the heard phonemes and seen lip movements are not synchronized.

We suggest that when the heard syllable is temporarily retained in the phonological loop (in STM), it can still be modulated by the subsequently-arriving seen lips movements (i.e., for those cases where the sound leads). Hence, combining the results of the present study with those from Soto-Faraco and Alsius's (2009) recent work, it would seem that there may, in fact, be multiple audiovisual interaction mechanisms in different domains, such as the CSTM for semantic representations, and the STM for phonological representations (see also Driver & Noesselt, 2008; Soto-Faraco & Alsius, 2009). These short-term buffers may retain and integrate the information from different sensory modalities in order to construct coherent multisensory representations suitable for further processing. Consequently, these short-term buffers (for more complex stimuli) may serve to widen the temporal window of multisensory integration beyond that seen in previous behavioral studies that have so often used simple light and sound stimuli.

7.1. A possible framework of multisensory integration

Previously, researchers have categorized the variables that influence audiovisual interactions into *structural* and *cognitive* factors (Radeau & Bertelson, 1977; Spence, 2007; Welch & Warren, 1980), which are thought to jointly determine the final percept (see Welch, 1999; Welch & Warren, 1980). Structural (or stimulus-driven) factors include such things as spatiotemporal coincidence and the intensity of the stimuli. Cognitive factors, on the other hand, are those that depend on prior knowledge or top-down control, and include the semantic congruency between the stimuli (as tested in the present study). Cognitive factors can modulate audiovisual integration, as has been demonstrated by research on the “unity assumption” (or “unity effect”; i.e., the belief that what is seen and heard belong together; see Vatakis & Spence, 2007; Welch, 1999; Welch & Warren, 1980). Accordingly, when the congruent semantic nature of auditory and visual inputs results in their being integrated together, any incongruent properties should be overridden and go unnoticed, such as that the spatiotemporal window is wider for semantically-congruent stimuli than for semantically-incongruent stimuli (Jackson, 1953; Parise & Spence, 2009; Vatakis & Spence, 2007). Recently, however, Spence (2007) argued that structural and cognitive factors are perhaps not as easily distinguished as has often been asserted. For example, although spatial and temporal coincidence are generally considered to be structural factors, it is important to note that they can also serve as cognitive cues to bind visual and auditory events as well (Arnold, Johnston, & Nishida, 2005). Furthermore, as the temporal windows in which simultaneity is perceived and in which the semantic congruency of audiovisual stimuli has been shown to influence multisensory integration can be dissociated (see the present study; Soto-Faraco & Alsius, 2009), the underlying mechanisms should therefore not necessarily be considered as being unified (see also Driver & Noesselt, 2008, on this point). A new classification of factors and framework for crossmodal integration is therefore needed.

One possibility here is to try and differentiate those factors that influence multisensory interaction in terms of the well-established classification that has already been worked out in the visual modality: spatiotemporal factors (‘when’ and ‘where’, or ‘token’) versus identity factors (‘what’, or ‘type’). Similar cognitive frameworks that have attempted to dissociate these two factors have been proposed previously, such as embodied by the notion of *object files* (Kahneman, Treisman, & Gibbs, 1992), the *visual indexing model* (Pylyshyn, 1989, 2001), and the *type-token framework* (Bowman & Wyble, 2007; Driver & Vuilleumier, 2001; Kanwisher, 1987, 2001; Marcel, 1983b). There is also evidence from neuroanatomical studies to support the existence of separate ‘what’ and ‘where’ pathways (e.g., Mishkin, Ungerleider, & Macko, 1983; Ungerleider & Mishkin, 1982; further dissociated as ‘what’ and ‘how’, by Milner & Goodale, 1995). This dissociation between the ‘what’ and ‘where (or how)’ pathways has now been extended to both the auditory (Kaas & Hackett, 1999; Kubovy & van Valkenburg, 2001; Rauschecker & Tian, 2000; Romanski et al., 1999) and somatosensory modalities

(Chan & Newell, 2008; Dijkerman & de Haan, 2007; Reed, Klatzky, & Halgren, 2005) as well. The evidence that is currently available therefore suggests that there is a common organizational principle for information processing in the visual, auditory, and somatosensory modalities. Thus, the dissociation between spatiotemporal and identity processing may provide two basic interdependent mechanisms capable of subserving multisensory interactions in human information processing.

Concerning the physiological basis of multisensory interactions, several subcortical and cortical areas, such as the superior colliculus, the superior temporal sulcus (STS), and the middle temporal gyrus (MTG) have all been shown to be involved in constraining the integration of crossmodal events (see Driver & Noesselt, 2008; Stein & Meredith, 1993). Recent studies have further probed the multisensory interactions taking place in those brain areas traditionally considered to be “sensory-specific” (e.g., Noesselt et al., 2007; see Schroeder & Foxe, 2005, for a review). By measuring neural activity, researchers have been able to show that multisensory interactions start very soon after the stimulus onset (i.e., within 100 ms in certain cases, see Giard & Peronnet, 1999; Murray et al., 2005; Shams, Iwaki, Chawla, & Bhattacharya, 2005).

The above-mentioned studies, however, tended to focus on the spatiotemporal aspects of multisensory interactions. To date, only a limited number of such studies have attempted to investigate crossmodal semantic congruency effects as in behavioral studies (see Doehrmann & Naumer, 2008, for a recent review). For instance, Molholm et al. (2004) provided event-related potential (ERP) data to suggest that crossmodal semantic congruency effects occur around 400 ms after stimulus onset (and are associated with the N400 component; see also Schneider, Debener, Oostenveld, & Engel, 2008). To date, neuroimaging studies have found some candidate brain areas in which audiovisual semantic interaction have been shown to occur. For example, STS and MTG are both involved in audiovisual object integration; however they do not appear to be sensitive to audiovisual semantic congruency (see Beauchamp, Argall, Bodurka, Duyn, & Martin, 2004; Beauchamp, Lee, Argall, & Martin, 2004; Hein et al., 2007; though see Belardinelli et al., 2004; Schneider, Debener, et al., 2008). A few studies have reported that activation in the inferior frontal cortex (IFC) is more pronounced under conditions of semantically-incongruent stimulus presentation than under conditions of semantically congruent presentation (Belardinelli et al., 2004; Hein et al., 2007); however, the activation of IFC may reflect the cognitive control induced by perceiving conflicting information (see Doehrmann & Naumer, 2008). One of the most plausible brain areas discussed to date that may be involved in the processing of at least certain kinds of audiovisual semantic congruency is the perirhinal cortex. Neural activation in this area has been shown to be sensitive to categories of living versus non-living things (see Murray & Richmond, 2001; Taylor, Moss, Stamatakis, & Tyler, 2006; Tyler et al., 2004), which is a basic organizational dimension for concepts (Caramazza & Mahon, 2003; McCarthy & Warrington, 1988). Given that this structure receives information from unimodal sensory regions that represent object features (see Taylor

et al., 2006), it is worth considering whether it may serve as the afferent site from the ‘what’ pathways of each sensory modality, that facilitates the identification of multi-sensory objects in the world around us. Furthermore, it is also worth investigating in future research the functional role of the neural circuit of perirhinal cortex and associated regions; that is, whether the activation of such a connectivity can last for hundreds of milliseconds, which may serve as the neural basis of a buffer such as CSTM (e.g., Gazzaley, Rissman, & D’Esposito, 2004).

In conclusion, the results of the experiments reported here demonstrate that the identification of a picture can be modulated by the semantic congruency of any sound that happens to be presented at around the same time. The effect of audiovisual semantic congruency on picture identification performance occurs within a temporal window that extends for at least 300 ms after the presentation of the picture. We suggest that this audiovisual semantic congruency effect likely depends on a short-term buffer for semantic processing (e.g., CSTM), which enables a person to access the meaning of a picture rapidly and to retain it for a short period of time thereafter.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.cognition.2009.10.012](https://doi.org/10.1016/j.cognition.2009.10.012).

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